

Fin o'Flaherty

Site Reliability Engineer III

fin@oflaherty.is | [707-520-4669](tel:707-520-4669) | fin.oflaherty.is | [github/finn-e](https://github.com/finn-e) | [linkedin/oflah3rty](https://linkedin.com/in/oflah3rty)

Current Location: Kentucky, USA (*Highly Motivated for Relocation / Open to Global Opportunities*)

| PROFESSIONAL SUMMARY

Highly analytical and performance-driven Site Reliability Engineer III with over a decade of experience designing, automating, and scaling high-availability cloud and containerized infrastructure. Expert in multi-cloud engineering (AWS & Azure), GitOps workflows, and modern operating systems like Talos OS. Proven track record of reducing operational overhead, managing large-scale database systems, and executing flawless migrations from legacy infrastructure to cloud-native platforms. **Actively seeking relocation opportunities.**

| TECHNICAL SKILLS

Cloud Platforms: AWS, Azure

Container & Orchestration: Kubernetes, Talos OS, Helm, Docker, Rook/Ceph, Flannel, Traefik Ingress

Infrastructure as Code & Configuration: OpenTofu, Terraform, Ansible

CI/CD & Automation: GitHub Actions, Custom Python & PowerShell Scripting, Core-Infrastructure Automation

Monitoring & Observability: Prometheus, Grafana, Loki Stack, System-wide Logging Integration

Secrets Management: Bitwarden Secrets Manager, HashiCorp Vault

Databases: MSSQL (Database Management), PostgreSQL, Flask/Django CRUD Implementations

Operating Systems: Linux (Arch, Debian/Ubuntu Server platforms), Windows Server Management

Methodologies: Site Reliability Engineering (SRE), GitOps, High-Availability Architecture, Disaster Recovery

| PROFESSIONAL EXPERIENCE

Site Reliability Engineer III | Xerox | Lexington, KY / Remote

August 2024 - Present

- › **Engineer and maintain critical infrastructure** for a high-traffic SaaS Publishing Platform for Retail across dual-cloud environments (AWS and Azure).
- › **Architect and manage containerized infrastructure on AKS (Azure Kubernetes Service)**, leading the strategic migration of legacy services to modern cloud-native orchestration.
- › **Administer an extensive fleet of virtual Windows servers** along with associated enterprise networking and cloud infrastructure dependencies.
- › **Drive comprehensive Database Management operations utilizing MSSQL**, ensuring data integrity, performance tuning, and high availability for client-facing retail workloads.
- › **Develop idempotent automation workflows** to maintain system reliability, minimize configuration drift, and standardise multi-cloud cluster environments.

Administrative Systems Engineer II | Kentucky Geological Survey | Lexington, KY

February 2022 - August 2024

- › **Spearheaded the design and bare-metal implementation of an 18-node Kubernetes cluster** utilizing Talos OS, Flannel network fabric, and Traefik ingress, achieving 99.99% uptime for mission-critical tools.
- › **Engineered data storage tier using Rook/Ceph**, optimizing high-performance persistent volume provisioning for over 50 databases and 20 virtual servers.
- › **Implemented unified GitOps workflows** using OpenTofu, Ansible, Helm, and GitHub Actions, reducing deployment errors by 75% and speeding up feature delivery by 40%.
- › **Secured cluster secrets and delivery workflows** by integrating robust secrets management using Bitwarden Secrets Manager and HashiCorp Vault.
- › **Wrote tailored Python automation tooling** that streamlined IT operations, increasing internal productivity by 40% and saving 2 hours weekly per engineer.
- › **Designed a unified observability stack** integrating Prometheus, Grafana, and Loki with active Windows Server systems to establish total system transparency.
- › **Created a bulletproof disaster recovery plan**, guaranteeing a 30-minute Recovery Time Objective (RTO) across core research endpoints.

Software Developer & Systems Administrator | Phoenix Flow Control | Houston, TX

January 2014 - June 2021

- › **Architected and wrote an integrated proprietary system** tracking inventory, manufacturing, sales, and pricing that directly drove a 15% increase in company revenue.
- › **Transitioned legacy on-premises infrastructure to AWS**, cutting total operational overhead by 50% and vastly increasing infrastructure scaling thresholds.
- › **Adopted an early Infrastructure-as-Code (IaC) methodology**, dropping overall environment footprint expenses by 40% over a 5-year timeline.

PROFESSIONAL PROJECTS

Kubernetes Infrastructure Overhaul (Internal KGS Initiative)

- › **Challenge:** Overhaul aging infrastructure to support an evolving suite of modern, data-heavy research tools.
- › **Solution:** Provisioned an immutable Kubernetes environment driven by Talos OS and automated entirely via declarative CI/CD pipelines.
- › **Result:** Decreased service deployment timelines from hours to minutes, establishing the stable architectural baseline for the Data Preservation initiatives.
- › **Tech:** Talos OS, OpenTofu, Ansible, Helm, Traefik, Flannel, Rook/Ceph

Thin Sections Database, API, and Frontend

- › **Challenge:** Build a fast, highly accessible platform for visualizing and querying microscopic geological datasets.
- › **Solution:** Authored a high-throughput Python/Flask REST API paired with an optimized Vue.js frontend featuring client-side filtering and aggressive caching.
- › **Result:** System reliably handles 10,000+ daily requests with a 50ms average response time while serving over 5,000 registered users.
- › **Tech:** Python, Flask, Vue.js, Advanced Caching Modules

Data Preservation Initiative

- › **Challenge:** Build a resilient archiving and library data discovery platform from the ground up to support over 100,000 document records.
- › **Solution:** Engineered a robust Django backend coupled with a responsive Vue.js frontend, integrating a high-performance full-text query processing engine.
- › **Result:** Improved search accuracy by 85% and achieved a processing pipeline capability scaling up to 1,000 document ingestions per hour.
- › **Tech:** Python, Django, Vue.js, Full-Text Search Indices

EDUCATION

Master of Science in Geophysics

University of California - San Diego | San Diego, CA

Graduated: December 2013

- › Specialized in advanced seismic data processing, mathematical modeling, and algorithm development.

Bachelor of Science in Physics

University of Louisville | Louisville, KY

Graduated: August 2012

- › Dual-degree program graduate with Honors (3.8/4.0 GPA). Senior project in Quantum Computing Algorithms.

Bachelor of Arts in Psychology

University of Louisville | Louisville, KY

Graduated: August 2012

- › Specialized in cognitive psychology and human-computer interaction (HCI) concepts to guide user-centric system layout designs.

ADDITIONAL INFORMATION

Relocation Status: **HIGHLY MOTIVATED TO RELOCATE.** Open to international opportunities.

Languages: English (Native)

Hardware & Tinkering Interests: Custom Ergonomic Mechanical Keyboards (PCB Design, Firmware Customization, Hardware Prototyping)

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